User Stories — Smart Irrigation Alert Dashboard

Project: Smart Irrigation Alert Dashboard
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User Story 1: View Real-Time Soil Moisture Dashboard (US-01)

As a: Farmer / Farm Owner
I want to: View a centralized real-time dashboard displaying soil moisture percentages and ambient temperatures
So that: I can instantly assess the moisture status of each crop field.
Priority: High — **Story Points:** 5

Acceptance Criteria:

- ✓ Dashboard displays cards/rows for each active irrigation zone (e.g., Wheat, Tomato, Rice).
- ✓ Shows Zone ID, Name, Moisture (%), Temp, Status, and Last Updated timestamp.
- ✓ Moisture levels $< 30\%$ are highlighted with a distinct red "Critical" badge.
- ✓ Moisture 30%-60% shows amber "Optimal"; $> 60\%$ shows blue "Moist".
- ✓ Dashboard automatically refreshes or allows manual 1-click refresh without page reload.

User Story 2: Immediate Alert Notification on Critical Low Moisture (US-02)

As a: Farm Manager / Agronomist
I want to: Receive automated visual alerts and an exception banner when moisture drops below 30%
So that: I can turn on the irrigation pumps immediately before crops suffer water stress.
Priority: Critical — **Story Points:** 5

Acceptance Criteria:

- ✓ Alert banner is pinned to the top showing the count of zones in critical status.
- ✓ "Active Alerts" section lists all threshold breaches with timestamp and recommended volume.
- ✓ Alert state updates within 2 seconds of sensor data simulation.
- ✓ Alert can be acknowledged by the operator.

User Story 3: Add & Configure New Sensor Zones (US-03)

As a: Farm Administrator
I want to: Add a new sensor zone with custom parameters via an input form
So that: I can expand the irrigation monitoring system when new fields are added.
Priority: High — **Story Points:** 3

Acceptance Criteria:

- ✓ User-friendly form with fields: Zone Name, Crop Type, Initial Moisture, Temperature.
- ✓ Validation: Zone Name must be non-empty, moisture must be between 0–100%.
- ✓ On submission, the new zone immediately appears on the dashboard.
- ✓ System returns a clean REST response (HTTP 201 Created).

User Story 4: Search and Filter Zones on Dashboard (US-04)

As a: Agronomist managing a large farm
I want to: Search zones by name and filter by status (All, Critical, Normal, Moist)
So that: I can quickly isolate problematic zones without scrolling through dozens of fields.
Priority: Medium — **Story Points:** 3

Acceptance Criteria:

- ✓ Instant search bar that filters the table/cards as the user types.
- ✓ Dropdown filter with options: All Statuses, Critical, Optimal, Saturated.
- ✓ Displays a friendly "No matching zones found" state when yielding zero results.
- ✓ Instant clear-filter button to reset view.

User Story 5: Status Drill-Down & Historical Sensor Readings (US-05)

As a: Crop Consultant / Farmer
I want to: Click on any zone to open a detailed drill-down modal showing historical readings
So that: I can analyze trends over time and optimize water usage cycles.
Priority: Medium — **Story Points:** 4

Acceptance Criteria:

- ✓ Clicking any zone card or table row triggers a detailed drill-down view.
- ✓ Drill-down view presents the last 10 simulated readings with timestamps.
- ✓ Displays minimum, maximum, and average moisture for the selected zone.
- ✓ Provides a "Trigger Immediate Irrigation" button that updates levels in real-time.